Intrusion Detection Systems (IDS) And Traffic Analysis

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Executive Summary

This lab focused on setting up an intrusion detection system and exploring the operational behavior of IDS towards suspicious traffic. Key activities includes configuring snort for ids operation, testing a dos attack on owaspbw, observing and analysing the traffic behavior of the attack, validating the effectiveness of the ids towards the traffic floods and observing the tcp behavior on wireshark. The lab revealed that IDS serves as an important part of security mechanisms when building defense systems.

1.0 Introduction

An **Intrusion Detection System (IDS)** is a security tool that monitors network traffic or system activities to detect suspicious behavior such as hacking attempts, malware communication, port scanning, brute-force logins, or abnormal data transfers, then generates alerts for administrators to investigate. IDS works mainly by analyzing traffic patterns and packet contents using **signature-based detection** (matching known attack patterns) and **anomaly-based detection**(flagging unusual behavior compared to normal network activity). Traffic analysis is the process of capturing and inspecting network data to understand what is happening on the network by checking key details like source and destination IP addresses, ports, protocols (TCP, UDP, ICMP, DNS, HTTP), session behavior, packet frequency, and traffic volume, which helps identify threats like DoS attacks, data exfiltration, or unauthorized access.

2.0 Aim and objectives

The aim of this lab is to set up and configure an Intrusion Detection System (IDS) on a Linux machine, analyze network traffic against the OWASP Broken Web Applications VM IP, and identify potential security threat

The Objectives of this lab is to

- configure snort for ids operation
- test a dos attack on owaspbwa
- observe and analyse the traffic behavior of the attack
- validate the effectiveness of the ids towards the traffic floods
- observed the tcp behavior on wireshark

2.1 Tools

- Kali Linux
- Snort
- Owasp bwa
- Wireshark
- hping3

3.0 Methodology

3.1 Preparing the Environment

- The owaspbwa VM was powered and actively running with an ip address of **192.168.56.101**
- Snort was installed using the following command
- **`Sudo apt update && sudo apt install snort`**
- The linux interface of interest was **eth1: 192.168.56.103**

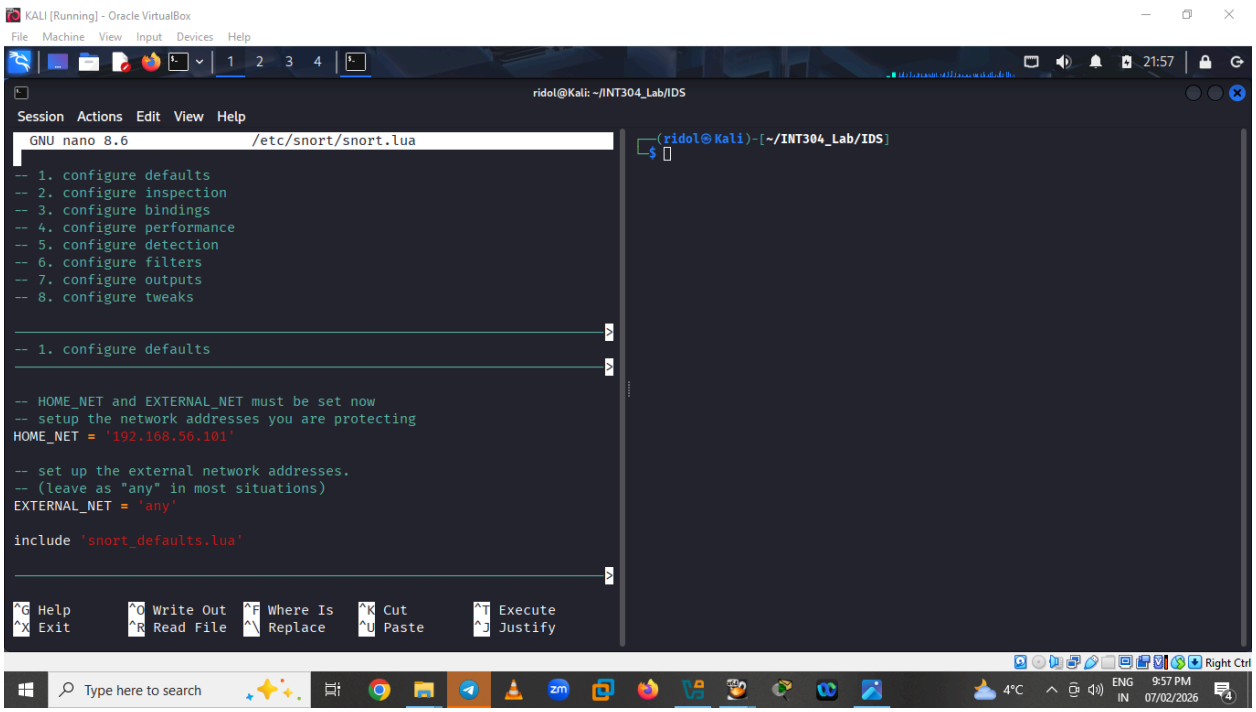

Fig 1: Evidence showing the snort configuration for the target 192.168.56.101 from a nano text editor

3.2 Setting Snort File Configuration

- The snort tools configuration file was open and edited using the following command
- **`sudo nano /etc/snort/snort.lua`**
- **`HOME_NET = '192.168.56.101'`**
- **`EXTERNAL_NET = 'any'`**

- **ips =**
- **{**
- **variables = default_variables,**
- **include = '/etc/snort/rules/local.rules',** -- custom rules path
- **}**
- The configuration was saved and exited

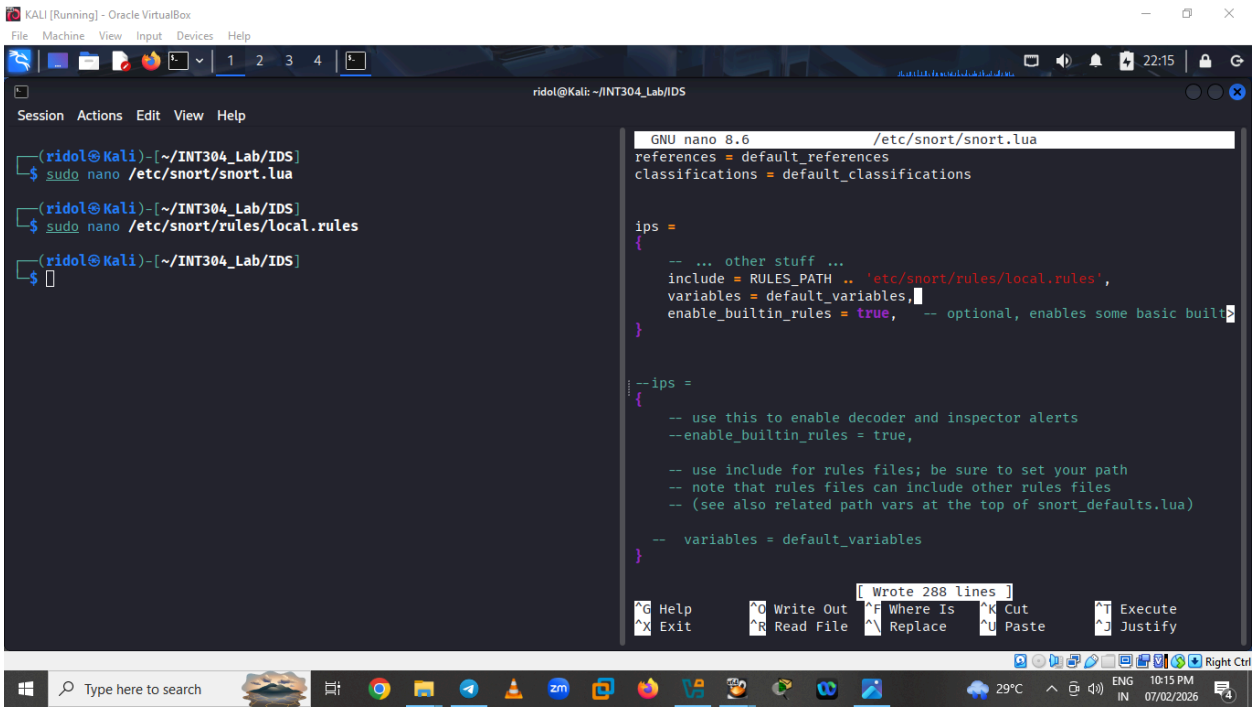

Fig 2: Evidence showing the snort configuration from a nano text editor

3.3 Setting Snort Rules Configuration

- The snort rule configuration file was open and edited using the following command
- **sudo nano /etc/snort/rules/local.rules**
- **alert tcp any any -> $HOME_NET 80 (msg:"Possible TCP SYN Flood Detected"; flow:stateless; flags:S; detection_filter: track by_dst, count 40, seconds 10; sid:1000001; rev:1; classtype:attempted-dos;)**

- The configuration was saved and exited
- Snort configuration was validated for error detection using the following command
- **`sudo snort -c /etc/snort/snort.lua -i eth1 -T`**

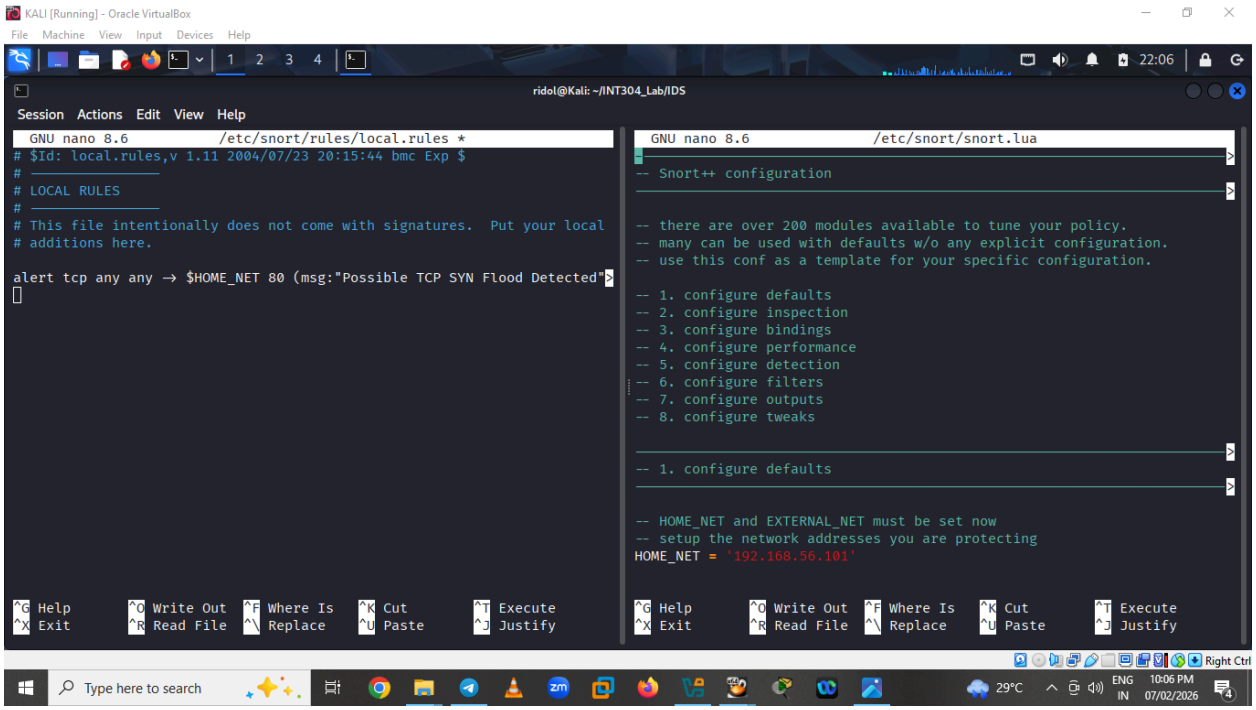

Fig 3: Evidence showing the snort rule configuration from a nano text editor

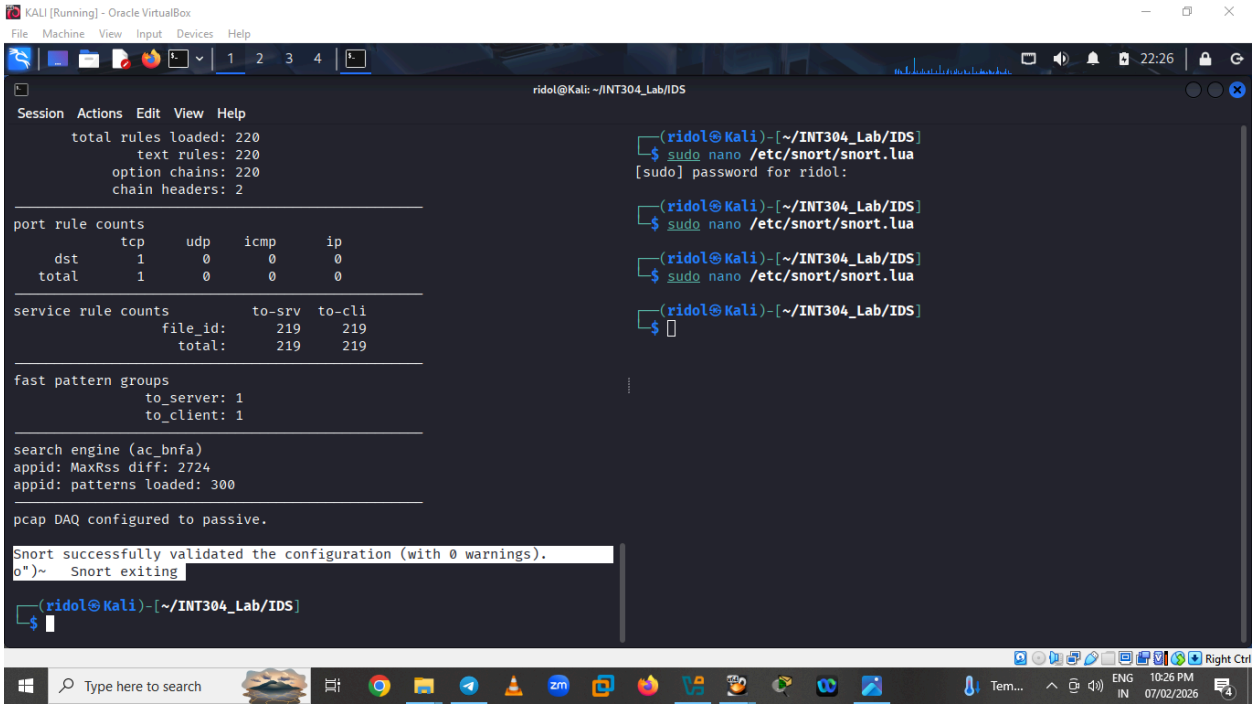

Fig 4: Evidence showing successful validation of the snort configuration

3.4 Starting Snort in IDS Mode

- Snort was launched using the following command
- **`sudo snort -c /etc/snort/snort.lua -i eth1 -A alert_fast`**
- Snort was successfully activated and actively waiting for traffic.

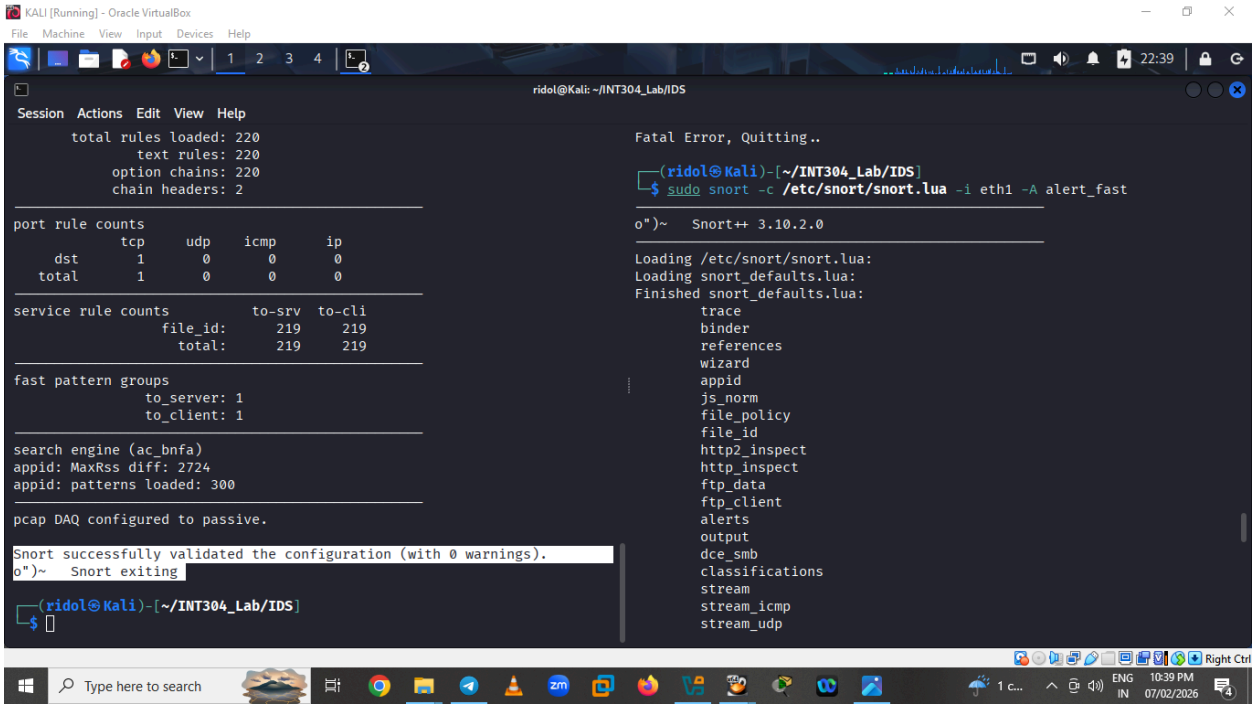

Fig 5: Evidence showing successful initiation of snort in IDS mode

3.5 Generating Malicious Network Traffic

- A malicious traffic was generated using hping3
- A **SYN** flood attack was launched on the OWASP VM using the following command
- **`sudo hping3 -S -p 80 --flood 192.168.56.101`**
- This triggered the snort IDS and a flood of packets were captured and flagged as malicious from the snort terminal.

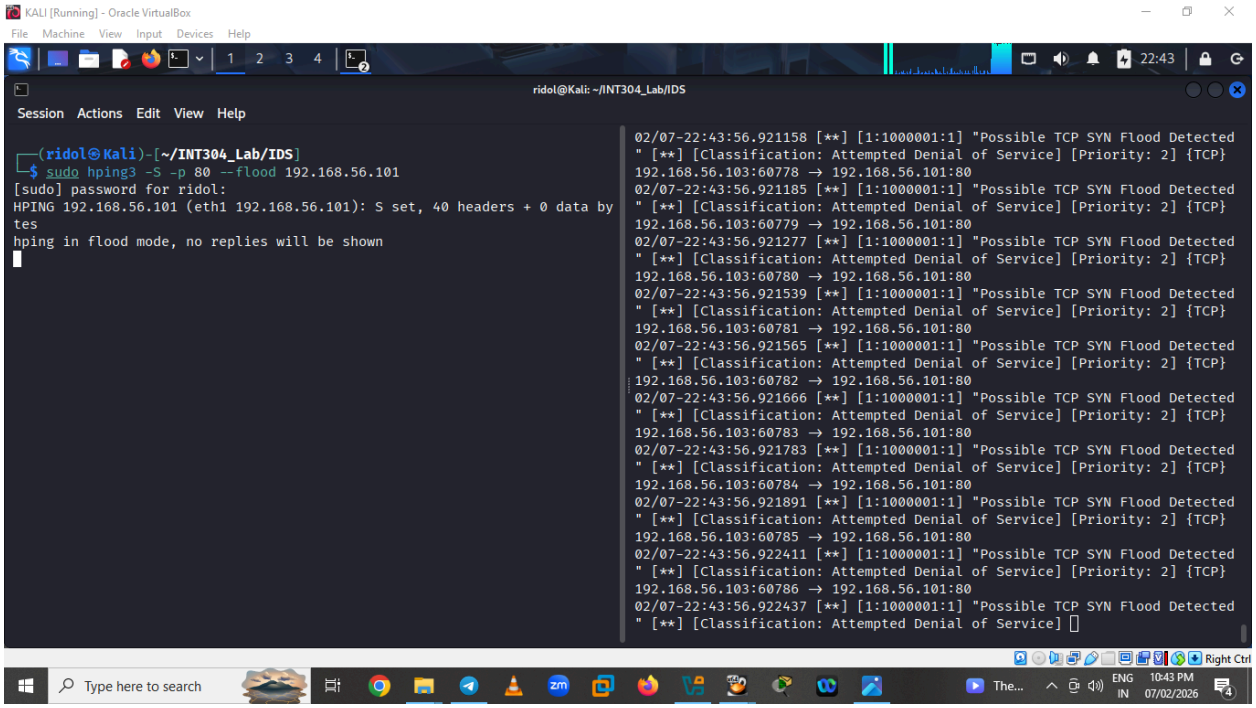

Fig 6: Evidence showing successful generation of malicious network traffic on Linux terminal

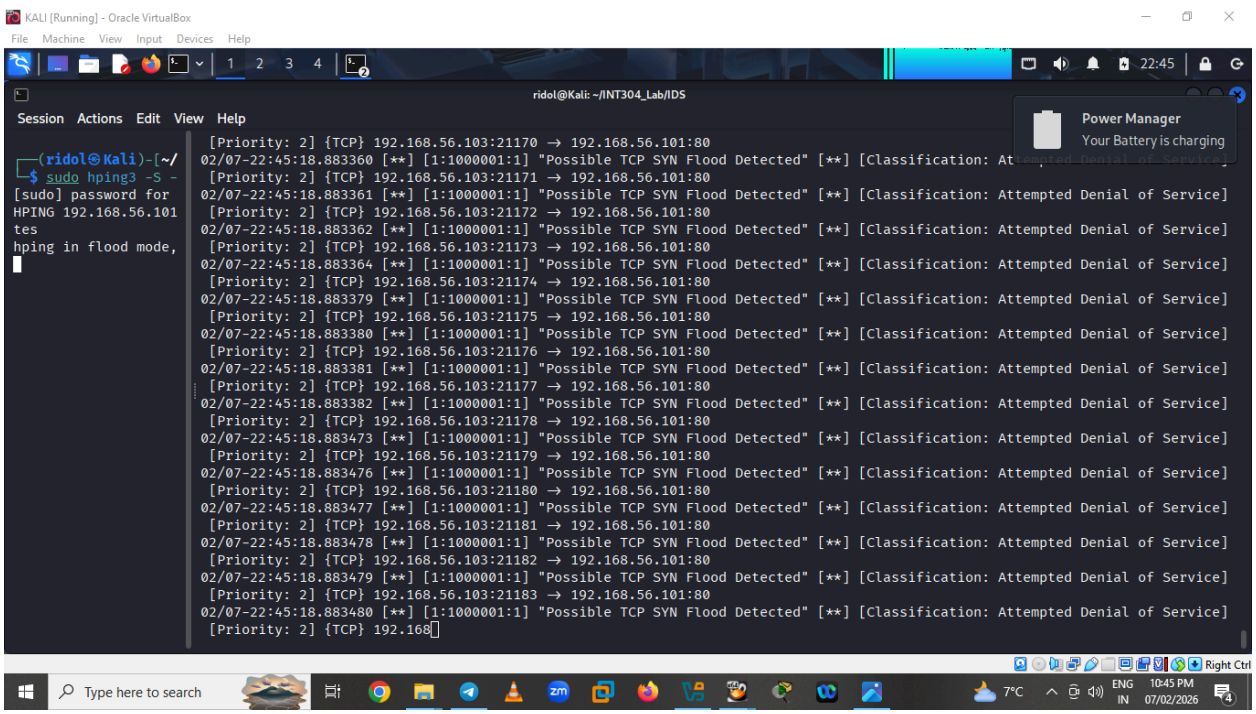

Fig 7: Evidence showing successful alert of malicious network traffic by snort on Linux terminal

3.6 Traffic Monitoring by Wireshark and Snort

- The wireshark was open with the command **`wireshark`** and the **eth1** interface was selected
- The filter **`tcp.flags.syn == 1 and tcp.flags.ack == 0`** was applied to filter out SYN flood packets only
- The **I/O** graph shows large amounts of bursty spikes with several tcp errors from the flooded packets.
- The snort was stopped and the traffic statistical summary was displayed.

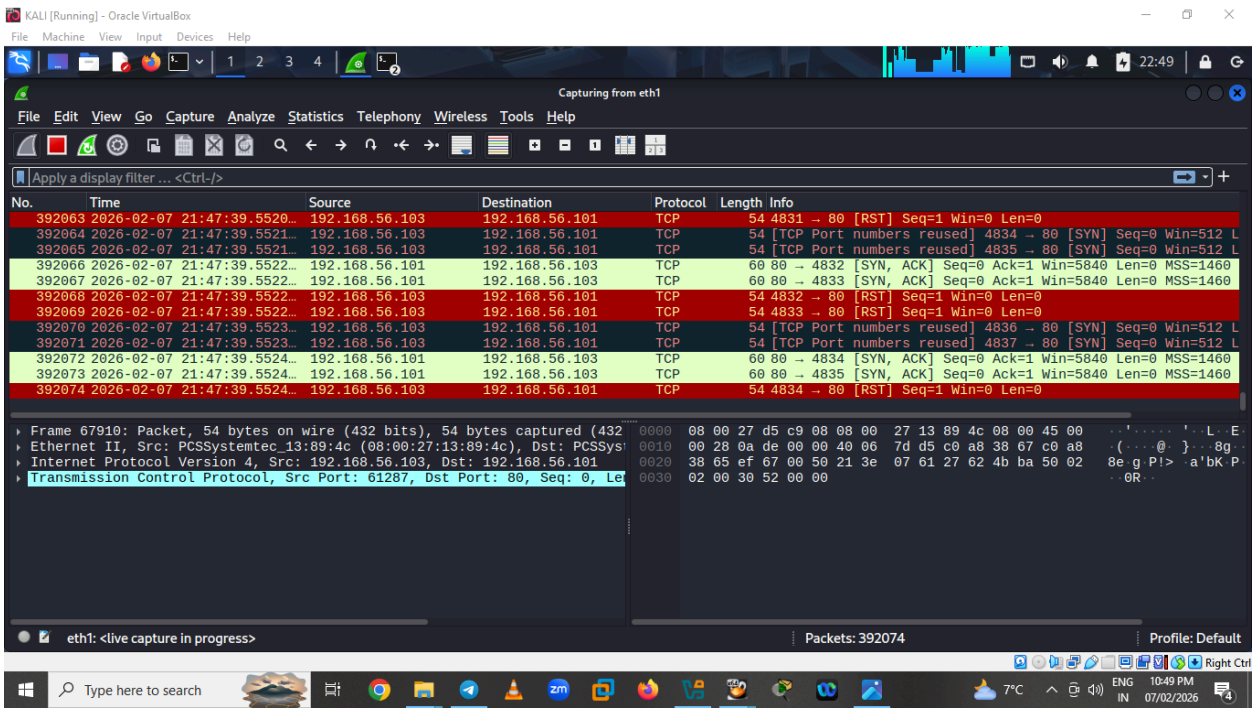

Fig 8: Evidence showing captured traffic on wireshark interface

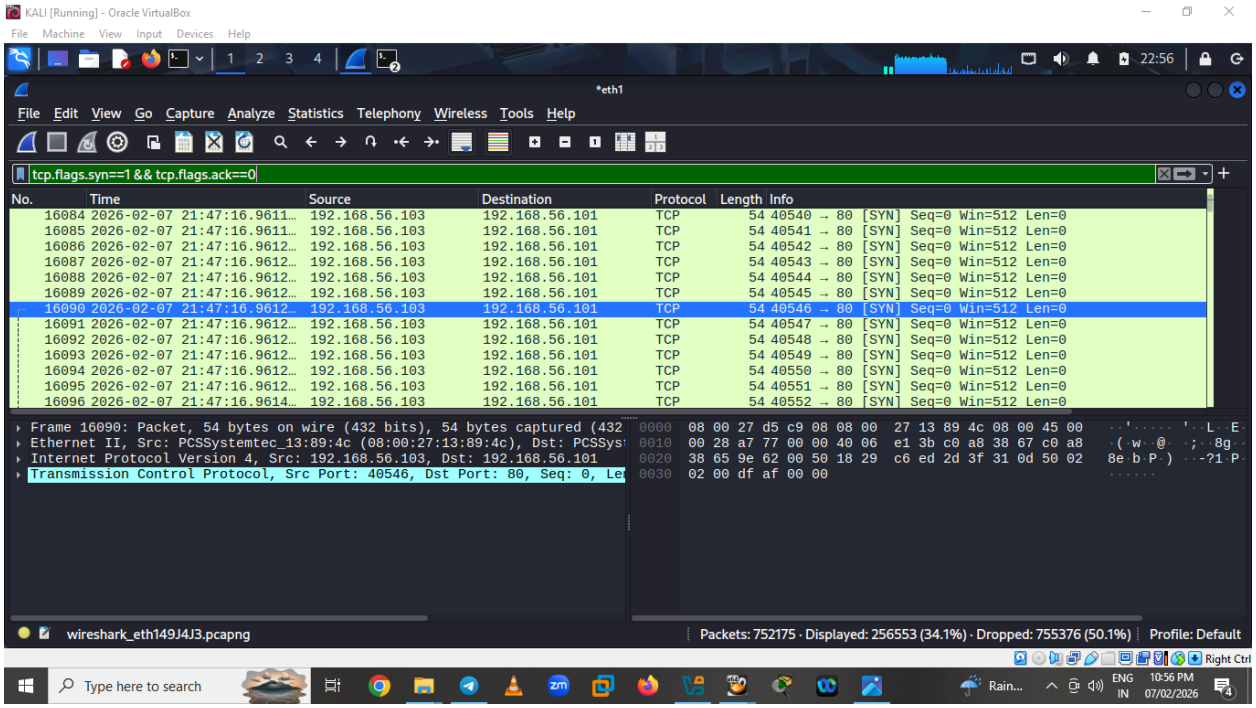

Fig 9: Evidence showing filtered SYN packets on wireshark interface

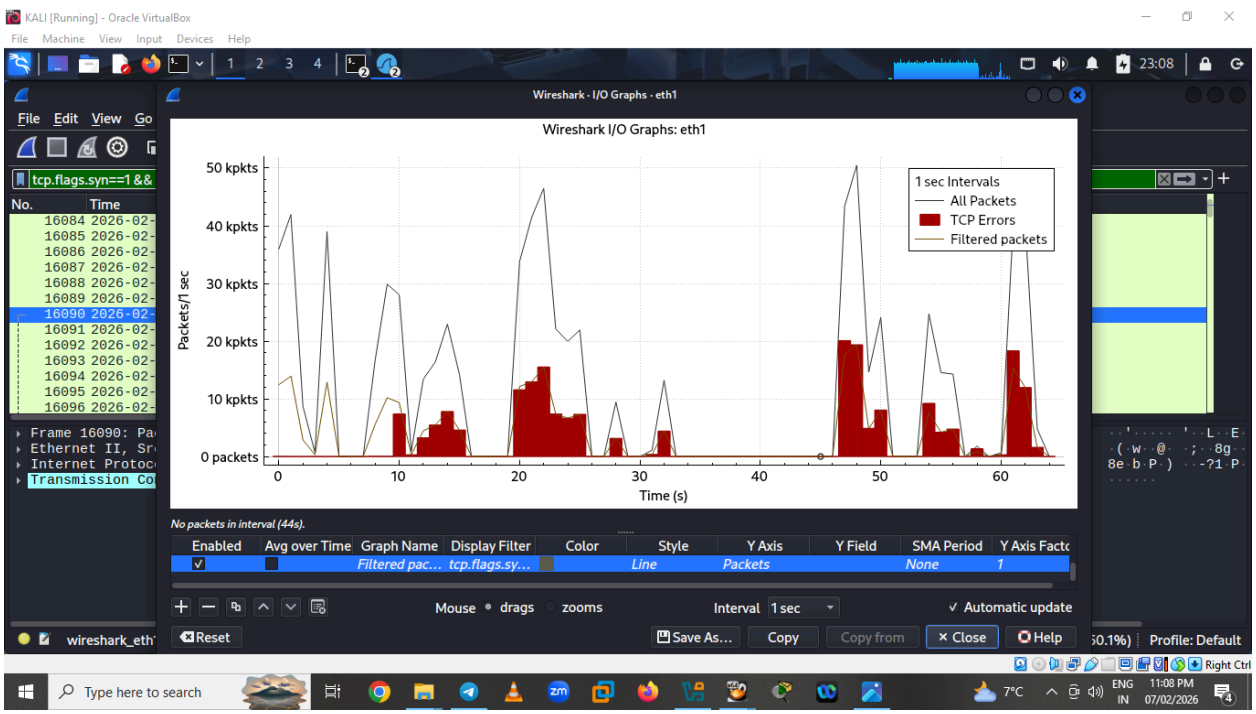

Fig 10: Evidence showing busty spikes of I/O graph from the captured packet on wireshark interface

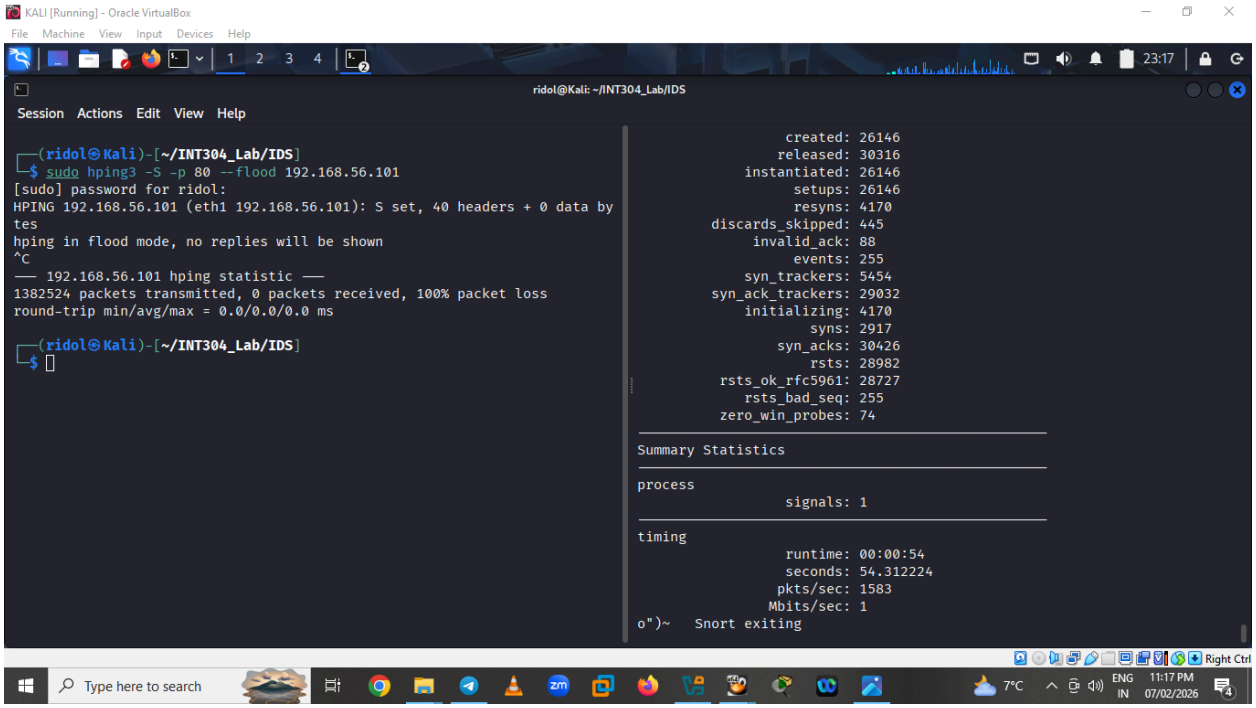

Fig 11: Evidence showing packets summary from snort on Linux interface

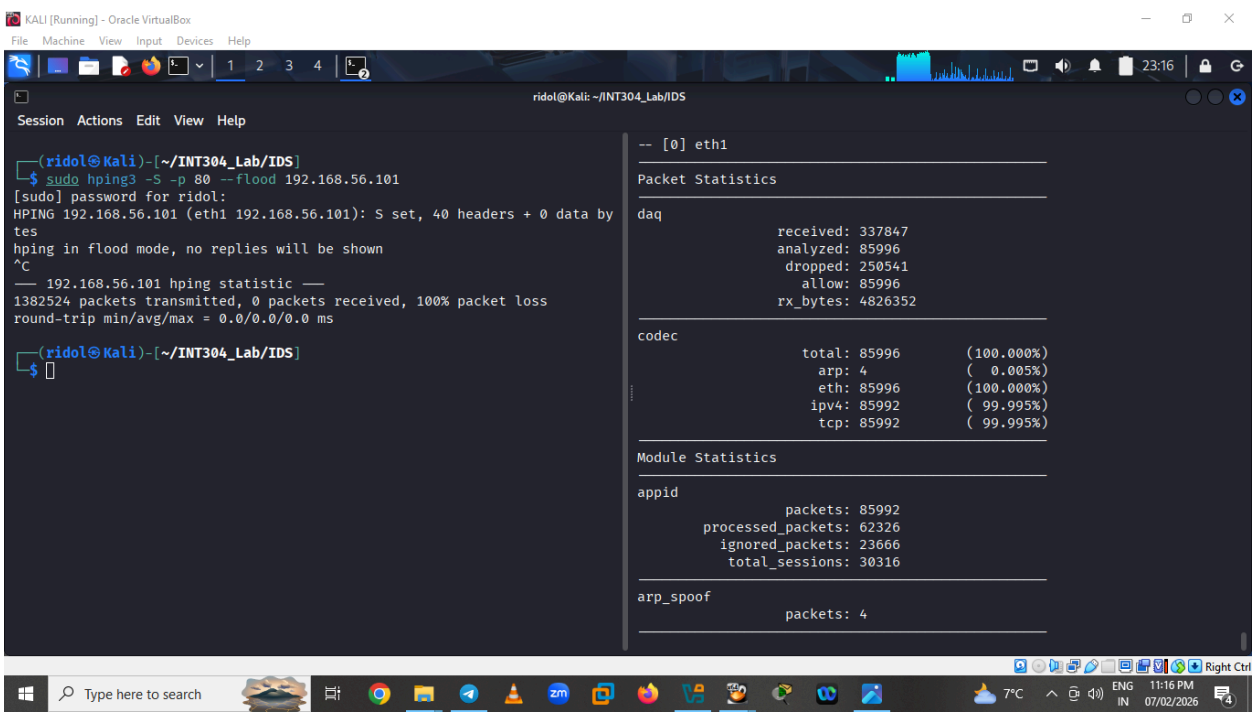

Fig 12: Evidence showing packets summary from snort on Linux interface

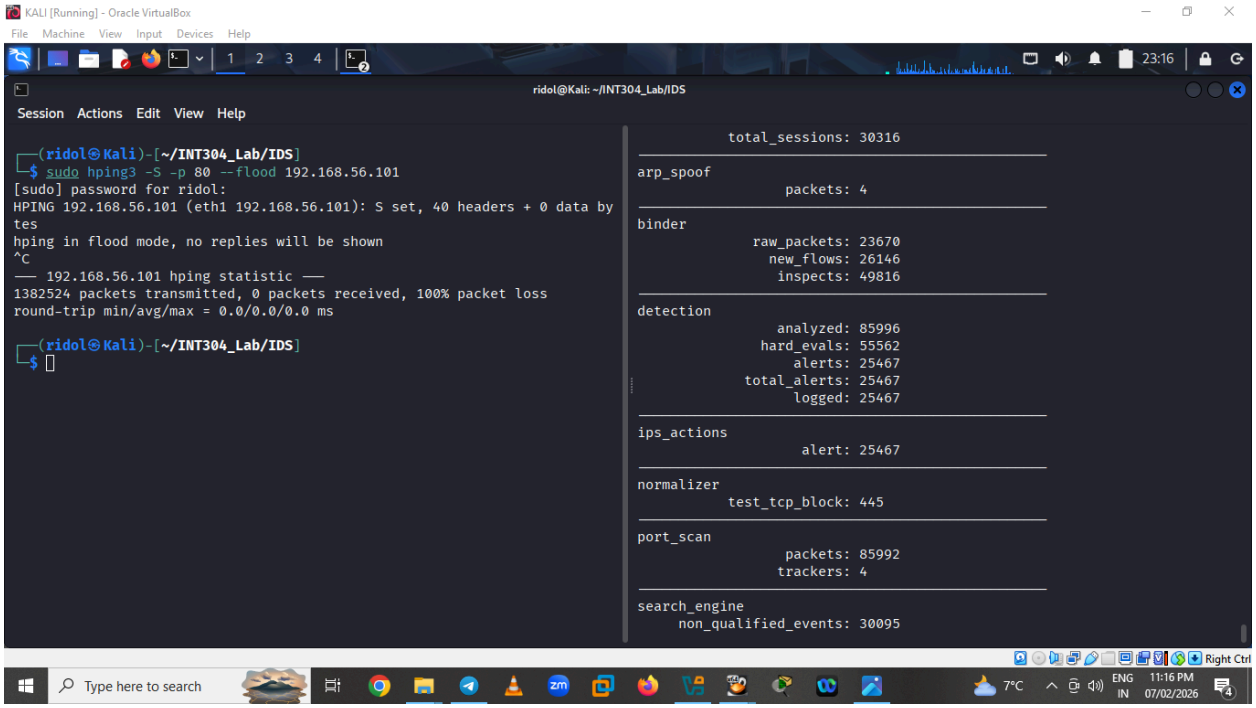

Fig 13: Evidence showing packets summary from snort on Linux interface

4.0 Results and Findings

Snort config file and rule file were configured to detect an anomalous traffic behavior from an attacker to owaspbwa target and flagged the behavior based on predefined rules such as packet rate and tcp handshakes. The SYN flood attack was observed and also confirmed from the snort terminal and wireshark refined filters and packets statistical I/O graph.

5.0 Conclusion and Recommendation

The flagged captured packets by snort indicates a successful IDS simulation . This acts as the first line of approach towards defense against cyberattacks such as DoS and DDoS on any network system by an intruder.

IDS isn’t a defense system , rather a real time notification trigger and insight towards an attack, mostly being more instrumental when combined with other preventive measures such as Intrusion Prevention Systems and Firewalls.